



EXAM PAPERS PRACTICE

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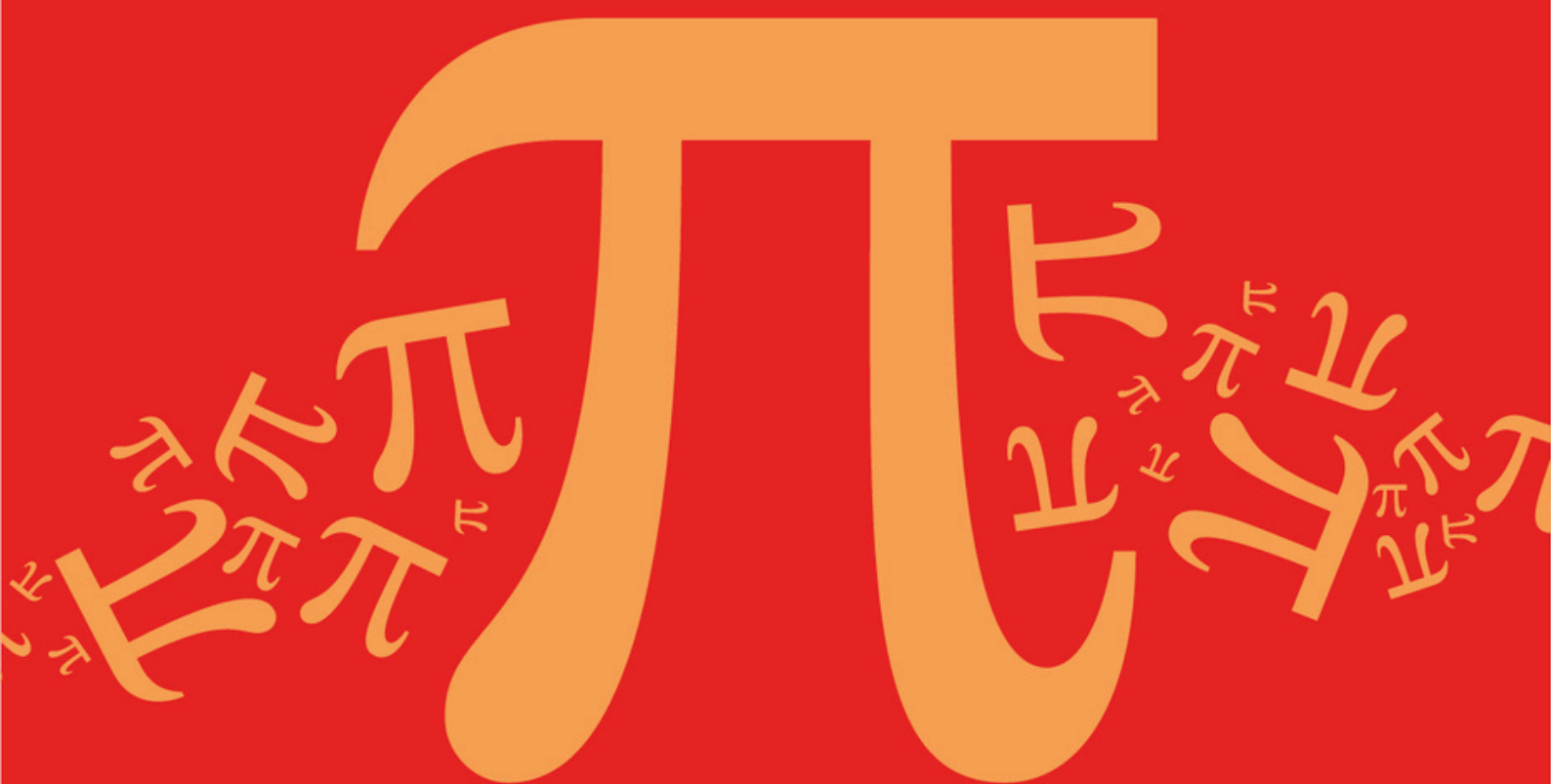
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

DB: Networks

Sub topic

DB4: Travelling Salesperson

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

DB: Networks

Sub topic

DB4: Travelling Salesperson

EXAM PAPERS PRACTICE

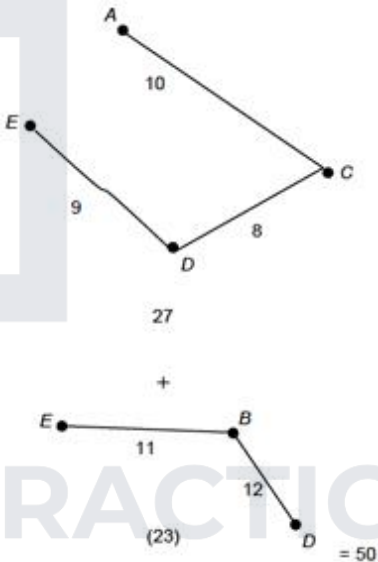
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the correct Hamiltonian cycle starting at A using the nearest neighbour algorithm	AO1.1a	M1	$A C D E B A$ $(10 + 8 + 9 + 11 + 15)$ $= 53$
	Correctly determines and states the upper bound	AO1.1b	A1	
(b)	Correctly determines a minimum spanning tree connecting A, D, E and C (or 27 seen)	AO1.1b	B1	
	Correctly identifies the two shortest arcs connected to B	AO1.1a	M1	
	Calculates the sum of 'their' minimum spanning tree and the two shortest arcs	AO1.1b	A1	
(c)	Infers correctly that the optimal Hamiltonian cycle has a weight between 50 and 53 minutes, and that there is currently no further information to suggest a route of less than 53 minutes exists as the lower bound is not a Hamiltonian cycle	AO2.2b	R1	The optimal time for a tour lies between 50 and 53 minutes, and the lower bound is not a Hamiltonian cycle, so we cannot be sure that a Hamiltonian cycle of less than 53 minutes exists.
(d)	Evaluates answers in parts (a) and (b) with reference to a plausible reason why Charlotte has taken longer than the typical tourist	AO3.5a	E1	The model only applies to a typical tourist. Charlotte takes longer than the typical tourist because she walks more slowly than the typical tourist.
(e)	Gives a plausible reason for why the times taken to walk between the monuments would likely be different during the	AO3.5b	E1	It is likely there will be less traffic at night, meaning that tourists travel time between each monument is likely to be less than the times during the morning.

evening			For example, they would not have to wait as long to walk across roads. OR At night there may be an increase in the number of pedestrians and it will therefore take longer to walk between the monuments.
Total 8 marks			

Q2.

(a) 58

B1 1

(b) E A C D B E
Or reverse

B1 1

(c) E A B D C E
Tour

M1

(8 10 15 10 23)
Visit all vertices

m1

Correct order

A1

= 66

If M0 scored, then 66 scores SC2

A1

CSO

4

(d) $\left. \begin{matrix} AB \\ BD \\ DC \end{matrix} \right\} (35)$

A spanning tree with 3 edges connecting A, B, C and D and 2 edges from E

M1

Correct m s t

A1

$$\left. \begin{array}{l} EA \\ EB \end{array} \right\} (17 = 52)$$

Correct edges from E

A1

52

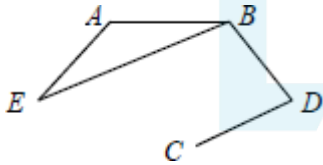
If M0 scored, then 52 scores SC2

A1

CSO

4

(e)



B1

Doesn't give a tour

Or other sensible conclusion

Eg: tour > 52 or 'doesn't give a solution'

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E1

2

[12]

Q3.

(a) 103

B1

1

(b) Tour

E1

May be improved

E1

2

(c) A C D F B E A
(11 10 15 15 24 27)

Tour, from A, visiting at least 4 other vertices, once only

M1

Visits all vertices

m1

Correct order

A1

= 102

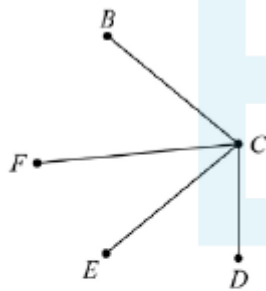
If M0 scored then 102 scores SC2

A1

CSO

4

(d)



+



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Spanning tree connecting B, C, D, E, F

AND

2 labelled edges from A

(for both, edges, not lengths, can be either listed or shown in diagram)

M1

Correct ST

A1

Correct edges from A

A1

= 77

If M0 scored then 77 scores SC2

A1

CSO

4

- (e) Min tour ≥ 77
 Allow their '77', provided '77' > 75

E1

1

[12]

Q4.

(a)

	A	B	C	D	E	F	G
A					8		
B					7		
C					13		
D					4		
E	8	7	13	4	—	10	19
F					10		
G					19		

5 correct values in an E 'line'

All correct

B1

B1

2

- (b) (i) **BADEFGCB**

Tour visiting at least 6 vertices

Visits all 7 vertices

M1

A1

Correct order from B

A1

80

B1

4

- (ii) **BADEFG E C A B**

Expansion of GC or CB

M1

Both correct

A1

2

- (iii) **76**

Minimum of 76 and their (b)(i)

B1F

1

(c) (i)

	①	②	④	③	⑤	⑥
	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>F</i>	<i>G</i>
<i>A</i>	-	2	6	4	16	27
<i>B</i>	②	-	8	3	15	26
<i>C</i>	⑥	8	-	10	23	32
<i>D</i>	4	③	10	-	12	23
<i>F</i>	16	15	23	⑫	-	20
<i>G</i>	27	26	32	23	⑳	-

Use of matrix form, 4+ numbers circled and 4+ parallel 'lines' crossed out

C added 4th

Any 5 values 'circled'

All correct values circled and lines crossed out, either as shown or as mirror image.
Order of vertices must be clearly shown.
Condone omission of line at *G*.

M1

A1

B1

A1

4

(ii) 43

For 43 seen, or for $2 + 6 + 3 + 12 + 20$

B1

$43 + (4 + 7)$

Their $43 + 2$ different edges from *E*

M1

= 54

SC 54 with no working 2 / 3

A1

3

(iii) 64

1

Must be written in symbols

2

[19]

(a)

1

Tour from A visiting at least 4 vertices

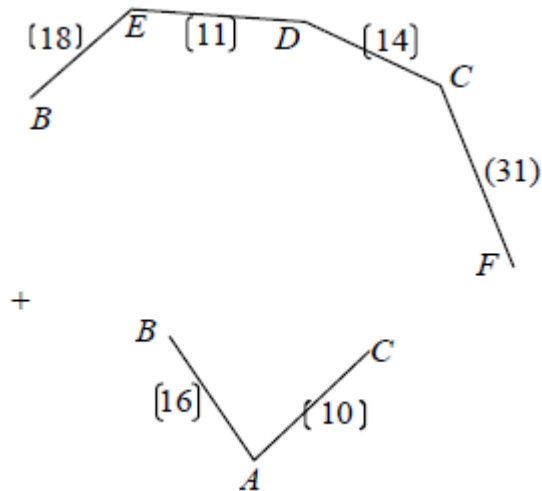
M1

Visits all vertices

Correct order from A

4

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Spanning tree + 2 different edges from A (ST must be edges using B, C, D, E, F not lengths, but condone two lengths from A, or 26)
 Diagram is not necessary in part (c)

Correct minimum spanning tree

Correct edges (not lengths) from A

M1

A1

A1

= 100

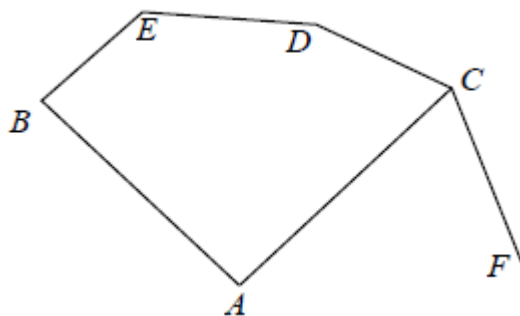
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B1

4

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(d)



Correctly labelled diagram

B1

Lower bound does not make a cycle

OE

AND tour > 100

Both, must be strict inequality

E1

Q6.

(a) 33

B1

1

(b) $B A E D C B$

Tour that visits all vertices

M1

Correct tour

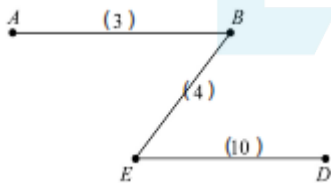
A1

= 41

B1

3

(c)



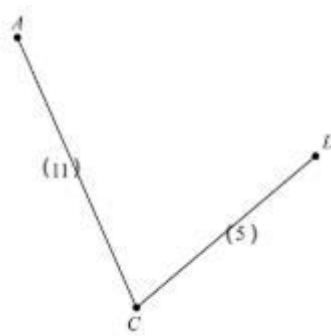
*Spanning tree without C
(either drawn or edges listed)*

and

2 different edges from C

(either drawn or edges listed)

M1



Correct MST

A1

Correct 2 edges from C

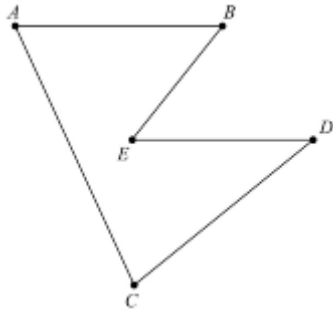
A1

= 33

B1

4

(d)



Correct network
Possibly earned in (c)

Optimal

OE

M1

A1

2

[10]

Q7.

(a) (i) $P U S R (= 40)$

E1

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Less than any other route
Or any one of $PQR = 50$, $PUQR = 45$,
 $PUR = 44$, $PUTSR = 54$ etc stated

E1

2

(ii)

	P	Q	R	S	T	U
P	-	25	40	24	26	14
Q	25	-	20	21	23	11
R	40	20	-	16	28	26
S	24	21	16	-	12	10
T	26	23	28	12	-	12
U	14	11	26	10	12	-

6+ correct either above or below diagonal

B1

All correct

B1

(b) (i) Q U S T P R Q

Tour visiting vertices once only (except start/finish vertex)

Visits all vertices

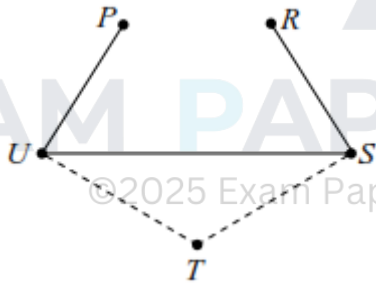
Correct order

= 119 (min)

(ii) Q U S T U P U S R Q

Any "expansion" of TP or PR from their (b)(i), PI

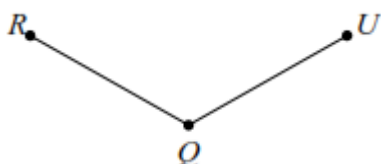
(c)



ST without Q (either drawn (vertices labelled) or edges listed)
and
2 different edges from Q (either drawn (vertices labelled) or edges listed)

either UT or TS in correct MST

4 edges in a labelled ST (must not include Q)



M1

m1

A1

B1

4

M1

A1

2

M1

A1

B1

Correct 2 edges from Q

A1

= 83

B1

5

[15]

Q8.

(a) (B E C D A B)
12(.0)

B1

1

(b) B D A C E B
Tour starts/finishes at B

M1

Visits B twice and all other vertices once

m1

Correct order

A1

If solution only on a matrix, then order of selection of vertices must be clearly shown

= 13.5

B1

4

(c) 12(.0)

Their min, condone writing 'part (a)' ft

B1F

1

(d) B A D E C B

Tour starts/finishes at B

M1

Visits B twice and all other vertices once

m1

Correct order

A1

If solution only on a matrix, then order of selection of vertices must be clearly shown

= 12.1

B1

4

[10]

Q9.

- (a) S T R I N G S
 64 70 82 80 82 72

Tour starting from any vertex

M1

Visits all other vertices only once

m1

Correct order

A1

= 450

B1

Note: If solution on a matrix then order of selection of vertices must be clearly shown

4

- (b) N G S T R I N
 Or reverse

Must have scored M2 in part (a)

B1F

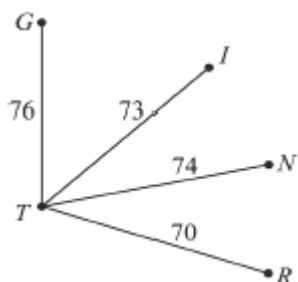
1

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- (c) Delete S

Clear method: spanning tree (edges or diagram, not just numbers) with one vertex deleted AND adding 2 edges from deleted vertex (condone double shortest edge from deleted vertex)

M1



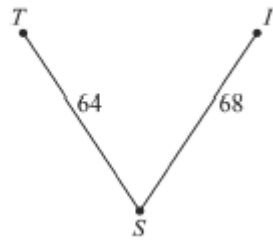
Spanning tree with 4 edges (may include S)

B1

Correct MST

A1

+



2 shortest from candidate's deleted vertex
(not shortest edge doubled)

A1F

= 425

SC 425 without earning first M1: 2/5

A1

5

[10]

Q10.

(a) (i) $2x - 4 < x + 6$
 $2x - 4 <$

M1

$\therefore x < 10$

CSO

AG

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A1

2

(ii) $2x - 4 < 3x - 7$ OE
 Allow any expression in matrix > 0

B1

$2x - 4 < 4x - 14$ OE
 Allow any expression in matrix > 0

B1

$\begin{pmatrix} x > 3 \\ x > 5 \end{pmatrix}$

2

(b) (i) $2x - 1 <$
 Condone $\leq$ for method mark only

M1

$$2x - 1 < 3x - 7$$

A1

$$2x - 1 < x + 8$$

A1

3

(ii) $\Rightarrow (x > 6)$

$$x < 9$$

Possibly earned in (b)(i)

B1

$$2x - 2 < 3x - 9$$

Condone $\leq$ for method mark only

M1

$$x > 7$$

A1

$$x = 8$$

B1

4

(iii) A 12 C 15 D 14 E 17 B 14 A

8x + 8 with their integer x

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M1

$$= 72$$

CAO (unsupported 72 scores M0A0)

A1

2

[13]

Q11.

(a) eg A B C D E F A

*Any tour ABA or better, any start vertex
but not revisiting a vertex
May be shown in a labelled diagram of a
cycle (eg triangle ABC)*

M1

With all vertices visited

May be shown in a labelled diagram of a cycle

A1

2

(b) (i) $F \quad D \quad C \quad A \quad B \quad E \quad F$
 (20) (15) (5) (25) (15) (15)
 (= 95) **AG**

Any tour, start/finish at F

M1

Visits all vertices

m1

Correct order

A1

3

If solution shown solely on matrix, then
 order of selection of vertices must be shown

(ii) Tour

"It's an answer", "a cycle", "it works"
 "it's possible ..."

E1

May be improved on

"Can't be worse", "not necessarily best",
 "could be improved"
 Not "can be improved"

E1

2

(c) $F \quad E \quad C \quad A \quad B \quad D \quad F$

Tour $FE(ABCD \text{ in any order with } B \text{ before } D)F$

M1

(30) (7) (5) (25) (11) (10)

Correct order

A1

= 88

B1

3

If solution shown solely on matrix, order
 of selection of vertices must be shown

[10]

Q12.

(a) (i) 40

B1
1

(ii) 40

B1
1

(b) $45 \leq T \leq 55$

B1
1

(c) (i)

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	–	20	38	35
<i>B</i>	20	–	18	15
<i>C</i>	38	18	–	33
<i>D</i>	35	15	33	–

3 indep correct

All correct

B1

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B1
2

(ii) *A B D C A*
20 15 33 38

Tour or visits all

M1

Correct order or their 33

A1

= 106

B1
3

(iii) *A B D B C B A*

Any expansion on (c)(ii)

M1

Correct

A1

2

[10]

Q13.

(a) (i) 130

$$\begin{pmatrix} T & P & V & B & C & T \\ 8 & 48 & 18 & 43 & 13 \end{pmatrix}$$

B1

1

(ii)

T	P	C	B	V	T
8	18	43	18	51	

*Tour (vertices or edges) starting from T
(Letters not numbers)*

M1

Visits all vertices starting from T

M1

Correct order

A1

= 138

B1

4

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(iii) A possible solution, eg tour
OE

E1

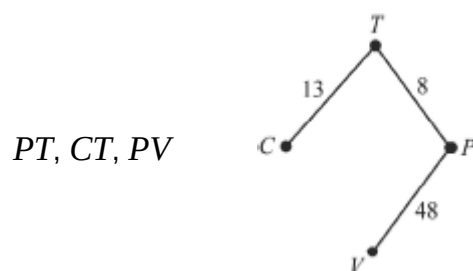
May be improved on

Allow '**can**' in this case as (i) < (ii) OE

E1

2

(b) (i)

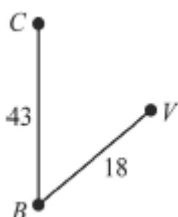


Spanning tree with 3 edges

Correct

M1

+ 2 shortest from B



A1

2 edges from B

Correct

m1

(Lower bound =) 130
CSO

A1

(ii) May not exist
OE

A1

5

Cannot be lowered
OE

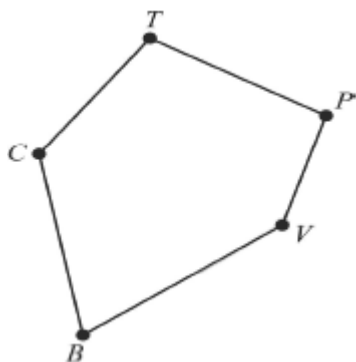
E1

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E1

2

(c)



B1

Tour or optimum or same as (a)(i)
Lower bound = Upper bound

E1

2

Q14.

- (a) $\begin{matrix} A & B & C & D & A \\ & 8 & 13 & 17 & 26 \end{matrix}$
4 numbers (either part)

M1

$= 64$

A1

2

- (b) $\begin{matrix} A & D & C & B & A \\ & 11 & 18 & 9 & 14 \end{matrix}$

$= 52$

A1

1

- (c) $\begin{matrix} A & C & B & D & A \\ & 6 & 9 & 25 & 26 \end{matrix}$
Tour

Visits every vertex

M1

Correct order

M1

A1

$= 66\}$

B1

Alternative if matrix used

M1 3 numbers } all different rows

M1 4th number } and columns

A1 correct numbers

B1 66

4

- (d) 52 (their lowest of (a), (b), (c))
Allow "part (b)"

B1F

1

[8]

Q15.

(a) (i)

$G \rightarrow P \rightarrow A \rightarrow N \rightarrow R \rightarrow G$
 65 115 155 125 160
 Tour

Visits all places

Correct order

Total = 620

M1

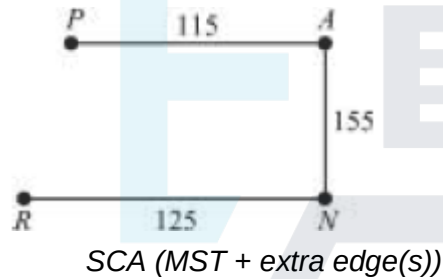
M1

A1

B1

4

(ii)

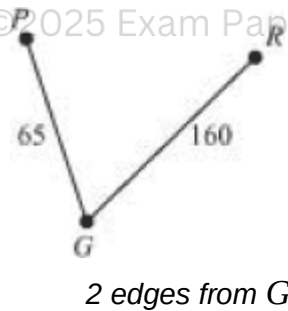


MST

M1

m1A1

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m1

$LB = 395 + 225 = 620$

A1

5

(iii) $T = 620$

Their (a)(ii) $\leq T \leq$ their (a)(i) where (a)(i) $\geq$ (a)(ii)

E1F

(b) (i) 92

		B1	1
(ii)	87		
		B1	1
(iii)	6		
		B1	1
(iv)	$n!$		
		B1	1
[14]			

Q16.

(a) (i) $L \quad N \quad O \quad L = 70$
 35 20 15

B1
1

$L \quad O \quad N \quad L = 95$
 35 40 25

B1
1

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(b) Any cycle

B1
1

(c) (i) $S \quad P \quad O \quad L \quad N \quad R \quad S$
 20 25 15 35 25 25
 Tour

M1

Every vertex

M1

All correct

A1
3

(ii) Tour

E1

May be improved

E1

2

(iii) *S* *R* *O* *L* *N* *P* *S*

30 17 15 35 21 20

Tour to every vertex with SR + N + P

M1

All correct

A1

= 138

B1

3

[11]

Q17.

(a) (i) 7

B1

1

(ii) 7

B1

1

(b) (i) Missing values
(PF 3) any 2 values correct

B1

$\left(OT3\frac{1}{4}\right)$ other 2 values correct

B1

2

(ii) FTPOMF
= $8\frac{1}{3}$ ISW

B1

1

(iii) FTMPOF
Tour

M1

Visits all vertices

Correct order

M1

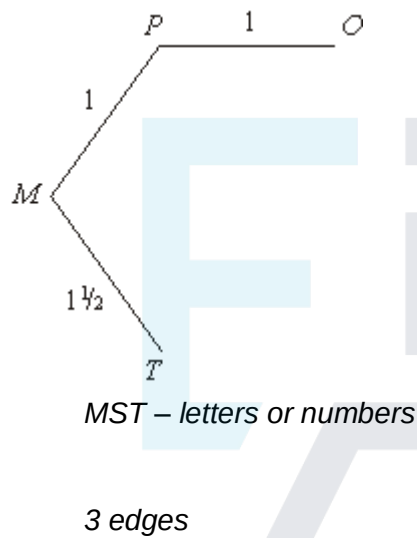
A1

= 7

B1

4

(iv) Delete F



M1

Correct

A1

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A1

Add $1\frac{2}{4} + 2$

Adding 2 edges from F

m1

= $6\frac{3}{4}$

SC $6\frac{3}{4}$ with no working $\frac{2}{5}$

A1

5

[14]

Q18.

(a) (i) A B C D E F A
6 values

M1

8 10 7 15 11 7

= 58

A1

2

(ii) $A \rightarrow C \rightarrow D \rightarrow F \rightarrow B \rightarrow E \rightarrow A$
 6 7 5 8 13 12

= 51

Tour starting and finishing at A

M1

Visits all vertices

M1

Correct order

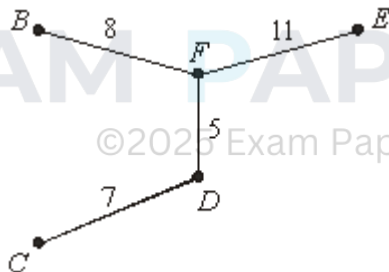
A1

B1

4

(b) Delete A

M1



SCA (MST plus 2 edges)

4 edges (not including A)

M1

Their MST + 6 (AC) + 7 (AF)

M1

Total = 44

A1

5

(c) $45 \leq T \leq 51$

M1

Use of inequalities

45

A1F

$$\text{Max } (45/\text{their}(b)) \leq T \leq \text{Min } (\text{their } (a))$$

51

A1F

3

[14]

Q19.

- (a) (i) $S \rightarrow R \rightarrow M \rightarrow B \rightarrow L \rightarrow S$
Adding 5 numbers

$$15 \quad 55 \quad 25 \quad 50 \quad 20$$

$$= 165$$

M1

A1

2

- (ii) $S \rightarrow R \rightarrow L \rightarrow B \rightarrow M \rightarrow S$
 $15 \quad 25 \quad 50 \quad 25 \quad 90$
Tour

Visits all vertices

M1

Correct order

M1

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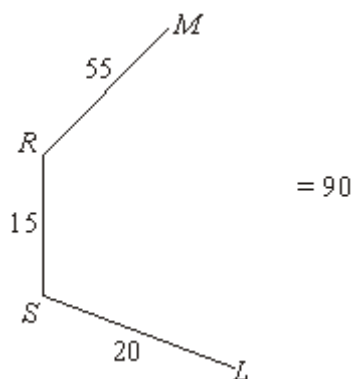
A1

4

$$= 205$$

B1

(b)



SCA (mst and 2 edges)

M1

3 edges

m1

Correct mst

A1

Choose 25, 50
(or BM, BL)

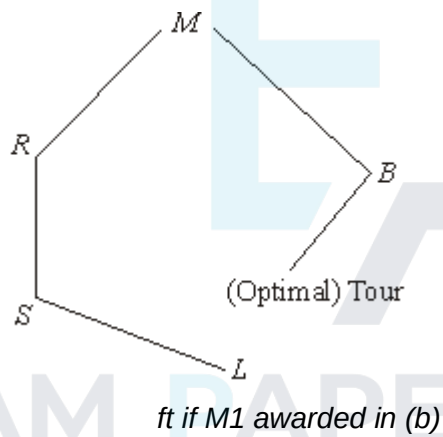
m1

Total 165

A1

5

(c)



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Either

B1F

E1

2

[13]

Q20.

(a) (i) $A G C V B A = 13 + 5 + 4 + 9 + 11 = 42$ (km)

B1

1

(ii) $A V C G B A$
 8 4 5 11 11
 Tour

M1

Visit all vertices

M1

Correct order

A1

$$= 39 \text{ (km)}$$

B1

4

- (b) (i) Delete A
MST – 3 edges

M1

$$(4 + 5 + 9) + (8 + 10) \\ \text{Correct MST} \quad \text{or } (4 + 5 + 9) + 16 = 34$$

A1

$$= 36 \text{ (km)}$$

B1

3

- (ii) Delete G
MST – 3 edges

M1

$$(4 + 9 + 8) + (5 + 6) \\ \text{Correct MST} \quad \text{or } (4 + 9 + 8) + 10 = 31$$

A1

$$= 32 \text{ (km)}$$

B1

3

- (c) $36 \leq T \leq 39$
 $34 \leq T \leq 39$
 Their (max (b)) $\leq T \leq$ Their(min(a))

B1FB1F

2

[13]

Q21.

- (a) A 7 B 8 C 5 D 10 E 3 A
 $= 33$

B1

1

- (b) Tour; may be improved

E1E1
2

(c) $A \rightarrow E \rightarrow C \rightarrow D \rightarrow B \rightarrow A$
 3 4 5 6 7
 Tour

visits all vertices

correct order

M1

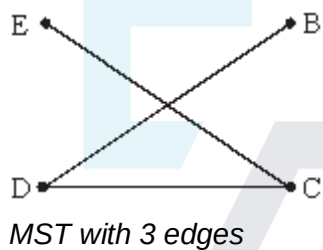
M1

A1

= 25

B1
4

(d) MST



= 4 + 5 + 6 = 15

M1

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A1

Add 3 + 7

M1

15 + 10 = 25

or 15 + 2 × 3 = 21

A1

4

(e) Cannot be lower than a lower bound

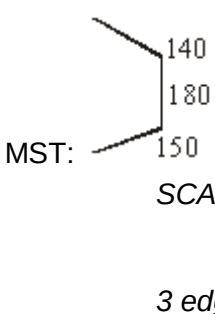
E1

1

[12]

Q22.

(a) $L A \rightarrow S B \rightarrow L V \rightarrow P S \rightarrow S D \rightarrow L A$
 90 140 180 150 185
 Tour

			M1*
	All visited		
			M1*
	* Independent		
	Correct order (must have both method marks)		
			A1
	Total 745		
			B1
		4	
(b)	Delete LA		
			M1
			A1
	= 470		
			A1
	LB = (their 470) + 90 + 185		
	or (their 470) + (2 × 90)		
			M1
	= 745		
	= 650		
			A1
		5	
(c)	Tour = 745		
	(b) ≤ T ≤ (a) or 650 ≤ Tour ≤ 745		
			B1F
		1	

[10]

Q23.

- (a) (i) $C \rightarrow D \rightarrow A \rightarrow F \rightarrow B \rightarrow E \rightarrow C$
 13 17 15 18 27 22
 Tour

M1

Visit every vertex

M1

F after A } dependant on starting from C

A1

All correct }

A1

= 112

SC: $13 + 17 + 15 + 18 + 27 + 22 = 112$ (4/5)

B1

5

(ii) A tour

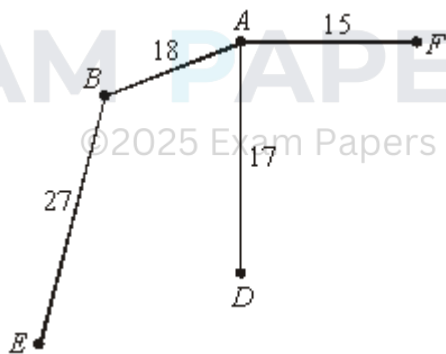
E1

May be improved

E1

2

(b) MST



Spanning tree

M1

4 edges (either AB or FB)

A1

77

CAO

A1

LB = (their 77) + (13×2)

(their 77) + 13 + 14

M1

$= 103$

$= 104$
CAO

A1

5

(c) $103 \leq T \leq 112$

or $104 \leq T \leq 112$
their (b) $\leq T \leq$ their (a)(i)

B1F

1

[13]

Q24.

(a)

	E	P	T
E	–	130	150
P	130	–	100
T	150	100	–

B3,2,1,0

3

(b) $C \rightarrow P \rightarrow T \rightarrow A \rightarrow E \rightarrow C$
40 100 145 95 90

Tour starts and finishes at C

M1

Visits every vertex

M1

All correct

A1

$= 470$

B1

4

(c) Actual route
 $C P C T C A E C$

M1

Once each A, E, P, T
 $2 (+ 2) C$

A1

2

[9]

Q25.

(a) (i) $A \rightarrow C \rightarrow B \rightarrow D \rightarrow A$

M1 M1 A1

Tour

M1

start and finish at A

M1

correct order

A1

$= 3!$

B1

4

(ii) $3 \times 2 \times 1 = 6$

or by listing

M1

SC 3

SC1

A1

2

(b) $n!$

factorial

M1 A1

2

[8]

Q26.

(a) (i) $S \rightarrow P \rightarrow O \rightarrow B \rightarrow L \rightarrow R \rightarrow S$
 10 15 5 25 15 15

M1

*a tour from S**
*visits all 5 vertices, independent**
**Independent*

M1

3

Correct order

A1

(ii) $R \rightarrow O \rightarrow B \rightarrow P \rightarrow S \rightarrow L \rightarrow R$
 7 5 20 10 25 15

M1

*a tour from R**
*visits all 5 vertices**
**Independent*

M1

Correct order

A1

Total = 82

(b) Tour

May be improved on

B1

4

E1

E1

2

(c) $S \rightarrow R \rightarrow O \rightarrow B \rightarrow L \rightarrow P \rightarrow S$
 20 7 5 25 11 10

A tour starting SR, or a tour that visits 5 vertices
in any order

M1

starts SRO

A1

starts SROB

A1

All correct

A1

Total = 78

B1

5

[14]

Q27.

(a) (i) $44 + 56 + 66 + 64 + 60 + 48$
 = 338

M1

A1 2

(ii) Tour

E1

Can be improved

E1 2

(b) Off $\rightarrow$ 1 $\rightarrow$ 3 $\rightarrow$ 5 $\rightarrow$ 4 $\rightarrow$ 2 $\rightarrow$ off
 44 50 62 60 62 52

*Start and finish at Office**

M1

*Visit all rooms**
**independent*

M1

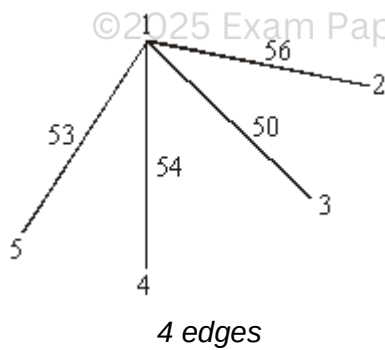
Correct order

A1

= 330

B1 4

(c) MST:



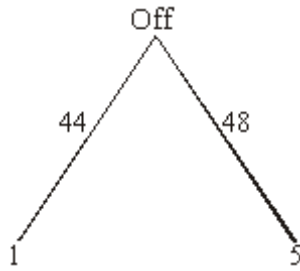
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M1 A1

= 213

B1

From Off:



M1

$$\therefore LB = 213 + 92$$

$$= 305$$

A1

5

$$\text{or } LB = 213 + 2 \times 44 \\ = 301$$

M1

A1

[13]

Q28.

(a) $M \rightarrow S \rightarrow C \rightarrow T \rightarrow P \rightarrow B \rightarrow M$

tour starts and finishes at M

M1

8 19 23 26 21 19
visits all vertices

for correct order

M1

A1

$$\text{Tour} = 116$$

(SC 97 scores 2/4)

B1

4

(b) MST is $B \rightarrow P \rightarrow S \rightarrow C \rightarrow T$

for spanning tree

M1

for four edges

A1

$$\text{MST} = 85$$

B1

$$\begin{aligned} \text{Minimum} &= (85) + (8 \times 2) \\ &= 101 \end{aligned}$$

$$\begin{aligned} \text{or } (85) + 8 \times 19 & \quad \text{their } (85) \\ &= 112 \end{aligned}$$

A1F

5

(c) $(101) \leq D \leq (116)$
 or $(112) \leq D \leq (116)$
 their (a) and (b)

B1F

1

[10]

Q29.

(a)

AT	7
TN	6
TC	8
TS	19
NB	20
BH	21
Min	81

B1

SCA Prim's

M1

For TS 4th

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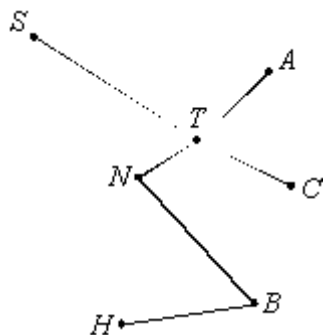
A1

For NB 5th

A1

Kruskal's or other Max 3/5

M1, B1, B1



B1

5

(b) (i) $MST (\text{No } A) = 74$

Their 81 - 7

B1F

$$\therefore LB = (74) + 2 \times 7$$

$$\text{or } (74) + 7 + 14$$

M1

3

$$= 88$$

$$= 95$$

A1F

- (ii) Join AT (twice)
Not complete tour
There are 4 single nodes
so impossible
or Join AT, AC

E2,1

2

[10]

Q30.

(a) $3x + 6 < x + 24$

B1

$$3x + 6 < 2x + 17$$

$$(-1 \text{ for } =)$$

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B1

$$3x + 6 < 5x$$

SC J $\rightarrow$ M can score B2 (ie - 1 MR)

B1

3

(b) $x = 4 < (2x - 3)$

M1 A1

$$x + 4 < (3x - 2)$$

$$OE$$

A1

3

(c) From (a)
 $x > 3$

B1

$x < 9$ B1

From (b)

$x > 7$

$(2x > 6 \text{ ignore})$

B1

$7 < x < 9$

$x = 8$

B1

Tour 104

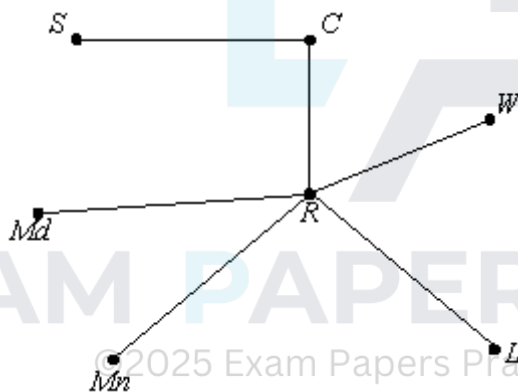
B1

5

[11]

Q31.

(a) MST [Mn = Milnrow; Md = Middleton]



s.c.a

M1

6 edges

M1

using C-S

A1

correct tree

A1

Length = $3 + 4 + 5 + 6 + 7 + 6$
 $= 31$

B1

5

(b) $R \rightarrow C \rightarrow S \rightarrow W \rightarrow L \rightarrow Mn$

s.c.a nn + final tour

M1

$\rightarrow Md \rightarrow R$

M1

complete cycle (8 letters) all correct

A1

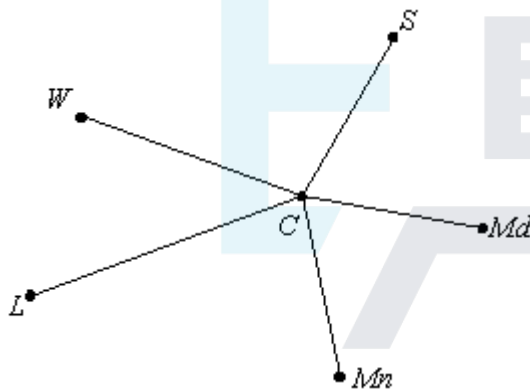
Upper bound = 57

B1

4

(c) Delete Rochdale

MST



M1

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A1

Lower bound = their (37) + (2 × 3)
allow their (37) + 3 + 4

M1

= 43

A1F

4

= 44

[13]

Q32.

(a) (i) $A B C D E F G A$
 = 131

s.c: 117 or route to G gets B1

M1

A1
2

(ii) Tour / initial solution

B1

May be improved

E1
2

(b) $A B D G F E C A$
start and finish at A

M1

visit all vertices

m1

correct order

A1

Total = 127

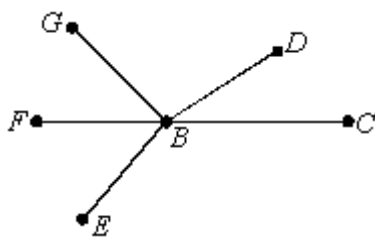
s.c: 110 or correct route to C gets B2 if returning and getting 220 then B1

B1
4

(c) Minimum spanning tree

M1

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five edges

A1

Minimum spanning tree = 87

Min = $87 + (2 \times 13)$

=113

or $87 + 13 + 14 = 114$

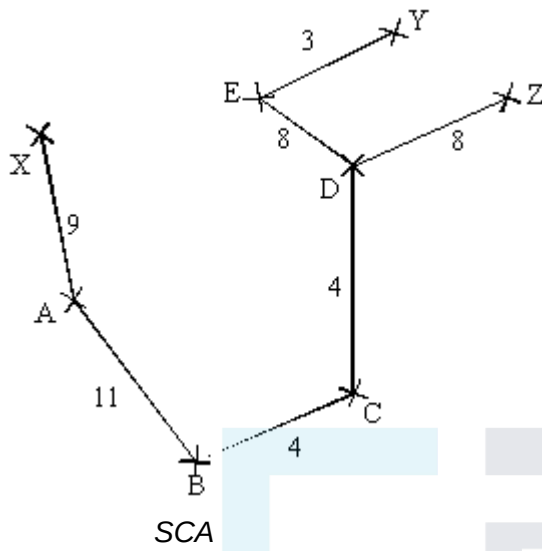
B1

f.t. from their spanning tree

M1
5

Q33.

(a)



MST = 47

cao

M1

A1

A1

3

(b) (i) $Z \rightarrow D \rightarrow C \rightarrow B \rightarrow A \rightarrow X \rightarrow E \rightarrow Y \rightarrow Z = 64$

M1A1

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2

(ii) $Y \rightarrow E \rightarrow D \rightarrow C \rightarrow B \rightarrow A \rightarrow X \rightarrow E \rightarrow Y \rightarrow Z \rightarrow Y = 78$

M1

M1

Using XEY

A1cao

3

(c) MST = 38

B1

tour = $38 + 14 + 9 = 61$

M1

Allow $38 + 2 \times 9 = 56$

A1f

3

(d) $61 \leq L \leq 64$

ft from their u.b. and l.b.

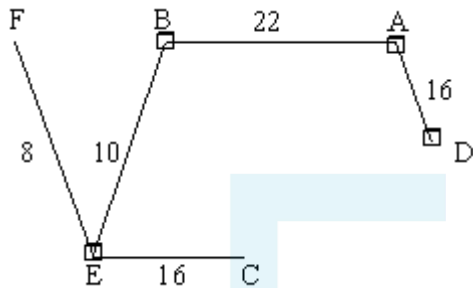
B1

1

[12]

Q34.

(a)



Prims/Krustals

[Either EC]

- 1 EE

M1

A3

4

(b) Del A: $8 + 10 + 16 + 32 + 38$
 $= 104$

M1 for either part

m1 for both part

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M1

M1

Del B = $8 + 16 + 16 + 26 + 22$
 $= 88$

A1

A1

Del C = $8 + 10 + 22 + 16 + 32$
 $= 88$

A1

Del D = $8 + 10 + 16 + 22 + 48$
 $= 104$

A1

6

[10]

Q35.

(a) $A \rightarrow D \rightarrow C \rightarrow B \rightarrow A$

M1A1

= 320 mins

A1

3

(b) Initial tour - may be improved on

E1

1

[4]

Q36.

University deleted	M	Li	S	B
Min spanning tree:	238	183	185	117
2 shortest:	56	56	61	122
connections	61	116	116	129
Lower bound:	355	355	362	368

Delete each university in turn

M1

Find the minimum spanning tree of the remainder

M1

All correct; -1 EE

A4

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Add shortest connections to deleted university

M1

ft minimum spanning trees; -1 EE

A3F

∴ Greatest lower bound = 368

Greatest lower bound; ft c's lower bounds

B1F

[11]

Q37.

LOVRMCL	175
OVRMCLO	175
COVRMLC	172
MROVLCM	175

M1M1

Correct order of letters -any using NN

A1×4

[6]

Q38.

(a) (i) *P S T...*

M1

... Q U W...

A1

... R V P

A1

Total $3 + 4 + 3 + 5 + 3 + 5 + 3 + 4 = 30$ miles

B1

4

(ii) e.g. *QT, RV, UW*

M1

TS

A1

QU, RW

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A1

Total $3 + 3 + 3 + 4 + 5 + 5 = 23$ miles

B1

4

(iii) Hamiltonian route $\geq$

M1

Min conn + lowest two links to P
 $= 23 + 3 + 4 = 30$.

A1

So the 30 found in (a) is best possible.

A1

3

(b) (i) The graph is K_8 with eight vertices of odd

M1

This needs at least four edges to make it Eulerian.

A1
2

- (ii) In order to add only 12 miles of roads

M1

look at the 4 roads of length 3 miles;
 PS QT RV UW .

A1

These do pair off the eight odd vertices and so repeating these roads will create an Eulerian graph.

A1
3

[16]

Q39.

- (a) (i) Nearest neighbour approach gives

M1

A E B C F G H D A

A1A1
3

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- (ii) It uses **all** the 1p links

B1
1

- (b) (i) Odd vertices A B C G

M1

Pairings $ABCG$: $2 + 2$
 AC BG : $(2 + 1) + 2$

A1

AG BC ; $(2 + 2) + 1$

A1

So repeat AB and CG

A1
4

- (ii) Repeat BC ($=1$),

B1

with message starting at A and finishing at G .

B1

2

[10]

Q40.

- (a) (i) There are some odd vertices

B1

1

- (ii) Odd vertices A, E, F, G

Pairings

AE/FG 12 + 10

AF/EG 10 + 11

M1

A1

AG/EF 13 + 9

A1

Minimum number 21

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B1

e.g. $ABFGFEFBADGEB$

M1

ECA

A1

6

- (b) (i) e.g. a "nearest neighbour" approach gives $ABFEDCA$

M1

with total $5 + 5 + 9 + 6 + 6 + 8 = 41$ but

A1

with G missed out. Could detour EGD to include G making total 50

M1

(Any sensible approach for Hamiltonian or non-Hamiltonian route of length ≤ 50 is acceptable)

A1

4

[11]

Q41.

(a) e.g. by Prim's algorithm:

M1

AC: 100

AD: 110

DF: 140

FE: 120

FB: 160

A1

A1

A1

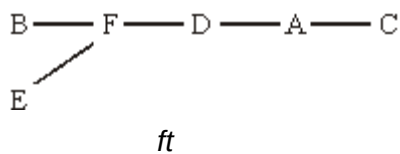
Total: 630 metres

B1

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5

(b)



M1 A1ft

2

(c) Need to clear *EB* and *BC*

Or, harder to spot, BC and DE

[N.B. EC is not a path in original]

M! A1

So length = 830

Or 820 respectively

B1

3

[10]

Q42.

(a) Odd vertices $P Q R U$.

B1

Pairings

Minimum extra

PQ RU

$2, 1 + 2 \Rightarrow 5$

M1

PR QU

$2 + 2, 2 + 2 \Rightarrow 8$

A1

PU QR

$3 + 2, 2 \Rightarrow 7$

A1

Therefore minimum closed Eularian walk is obtained by repeating PQ and RSU to give a total of $34 + 5 = 39$

A1

5

(b) $SRQPTVUS$

M1 A1 A1

$1 + 2 + 2 + 4 + 2 + 3 + 2 = 16$

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B1

4

(c) (i) Two edges at U must be used.

B1

(ii) In the reduced network the 'bottleneck' at S prevents a Hamiltonian cycle.

B1

(iii) (i) & (ii) imply that the length of a Hamiltonian cycle

M1

$\geq 4 + 3 + 1 + 2 + 2 + 2 + 2 = 16$

A1

4

(d) (i) Hamiltonian route clearly required.

M1

Shortest time = $16 + 20 \times 6 = 136$ seconds
ft (and allow 156)

A1ft 2

- (ii) From (a) the shortest route uses 16 edges and takes time = $39 + 20 \times$ (no of times route passes through a centre)

ft

M1

A1ft

= $39 + 20 \times 15 = 339$ seconds
(allow 359)

A1 3

[18]

Q43.

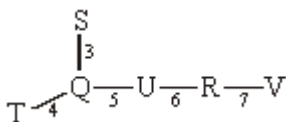
- (a) $P \rightarrow S \rightarrow Q \rightarrow T \rightarrow U \rightarrow R \rightarrow V \rightarrow P$

M1

3 3 4 7 6 7 4
any appropriate method

A1 A1 A1 4

- (b)



M1 A1

length 25

A1 3

- (c) L , shortest route $\leq$ answer in (a) = 34

B1

min connector of rest + 2 shortest

M1 A1

links to P = $25 + 3 + 4 = 32$

$\therefore L \geq 32$

A1

4

[11]

Q44.

(a) *DFGEJIHD*

M1 A1 A1

Length 53

B1

4

(b) *EG, EJ, HI, IJ, FG*

Kruskal or Prim

M1 A1 A1

Length 40

B1



B1

5

(c) Minimum = 40 + 6 + 7

M1 A1

So 53 best possible

A1

3

[12]

Q45.

(a) Prim (matrix or graphical) or Kruskal,

M1

giving AB; BD; CD; DE

A2

Total length 155 miles

–	45	60	50	90
45	–	67	25	80
60	67	–	50	70
50	25	50	–	35

90 80 70 **35** –

B1

(b) e.g. (nearest neighbour)

M1A2

A B D E C A

$$25 + 25 + 35 + 70 + 60 = 235$$

A1

(c)

–	45	60	50
45	–	67	25
60	67	–	50
50	25	50	–
(90	80	70	35)

M1

A1 45,25,50

M1

$$45 + 25 + 50 + 35 + 70 = 225$$

A1 35 and 70 (allow 2×35)

(d) Any tour is an upper bound. 235

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M1A1

(M0 for $2 \times$ min connector)

14

[14]

Examiner reports

Q2.

Several students ignored the paper rubric and presented their working on the diagram in the question. The first three parts of the question were usually answered very well. Most knew the required algorithm for part (d), but while presentation is now significantly improved from that of past years, a significant minority still lost marks by not clearly detailing which edges were involved. Lettered labelling is required, not just numbers. For the last part, those who had the required two parts for the diagram remembered to join them. Surprisingly few students realised that the fact that this diagram was not that of a tour was significant, with many students stating that the route was the optimal solution (presumably because that is what the comment often involves)!

Q3.

A significant number of students failed to read the instructions on the title page and within every designated answer space, and used the diagram given in the text of the question.

- (a) This part was almost invariably answered correctly.
- (b) This was the first question to cause significant difficulty. The most common misconception was to believe that the answer in the question could certainly be improved. Many students failed to state the obvious but necessary fact that the answer was in fact the time for a tour.
- (c) This part was usually answered correctly. However, a common error was to fail to complete the tour by returning to A . Those who used a redrawn table often failed to make clear the order in which edges were to be chosen.
- (d) The method is known and understood by most students. However, there is a widespread tendency to deal merely in numbers and to not clearly identify the edges being used. This results in the loss of a significant number of marks. Many students thought they needed a tour of $BCDEF$.
- (e) The reason was apparently widely known but the ability to express it clearly in comprehensible words was often lacking.

Q4.

There were very few errors in part (a) which was set in order to give candidates the opportunity to 'see' the context of the question.

In part (b)(i) the candidates who elected to write their tour down, vertex by vertex, underneath the table usually earned full marks. Those who elected to work entirely in the table often failed to make clear the order in which vertices had been selected and thus that they had actually used the nearest neighbour algorithm.

Despite answering part (b)(i) correctly and the hint in part (a), a surprising number of candidates merely repeated their tour from part (b)(i) for part (b)(ii).

Part (b)(iii) was usually correct.

Part (c)(i) was poorly answered. Many candidates simply did not present their working "on Table 2" and of those that did, few showed the order of selection of edges.

Part (c)(ii) caused even more problems. Many thought the required lower bound for the length of the tour would be the same as the length of the spanning tree just found. A larger number appeared to misread the requirement completely and now deleted a vertex from Table 2, going on to find a lower bound for a tour of the vertices in Table 2.

The last two parts of the question, part (c)(iii) and part (d), were quite well answered by the majority of candidates.

Q5.

There were very few errors in part (a).

In part (b), the students who elected to write their tour down, vertex by vertex, underneath the table usually earned full marks. Those who elected to work entirely in the table (notwithstanding the rubric as to where answers were to be written) often failed to make clear the order in which vertices had been selected and thus that they had actually used the nearest neighbour algorithm.

In part (c), many students knew what was required but very many did not show sufficient workings, and thus failed to earn full, or in some cases any, marks. Edges must be clearly identified to demonstrate clearly the method.

Very few students were able to provide the necessary explanation for part (d) often noting that the result was not a tour / cycle but not recognising the significance that this meant the tour would be greater than 100.

Q6.

This question proved to be very accessible to most candidates. The vast majority scored full marks on parts (a) and (b). The most common errors that did occur were simple slips - of notation or arithmetic. Presentation of the answer to part (b) was generally good.

The majority of candidates knew what was required for part (c). Presentation of this work has improved but there is still a minority of candidates that appear to think that the answer is the only thing of importance and demonstration of use of the algorithm unimportant.

Those successful with part (c), usually earned full marks for part (d).

Q7.

This question was a variation on a standard theme but it was accessible to most candidates who were able to attempt the early parts. The application of the classical problem to a practical context differentiated between candidates who fully understood the problem in part (b)(ii) but allowed weaker candidates to score good marks on parts (a) and (b)(i). In part (a), many candidates lost a mark by failing to explain why the route *PUSR* was the shortest, being content merely to verify that it was 40 minutes in extent. There were very few marks lost in filling the table.

Part (b) caused more problems. Some candidates failed to realise the significance of the words in part (b)(i): "... on **the table**..." and worked with the diagram. Many of those who worked with the table stated clearly only the final numerical answer and completely failed to explain clearly their method, which required a clear statement of the derived tour and evidence that the nearest neighbour algorithm had been used. Most candidates completely missed the point of part (b)(ii) and simply repeated their answer for the previous part, not realising the need to refer back to the context in the diagram.

Part (c) was a problem for candidates who failed to show a clear method of obtaining a

lower bound. This was quite a simple problem but candidates lost marks by not writing down edges or even a diagram identifying the edges. Many candidates found the correct answer of 83 but for many candidates this was at the end of a series of sums of unidentified numbers. For full marks, the algorithm used needed to be clear with the spanning tree edges and the two shortest edges from Q clearly identified. This could be done either by clear diagrams or text.

Q8.

This was another high scoring question with only the weakest failing to score full marks.

Q9.

Part (a) This was possibly the weakest area of basic knowledge with many failing to take their 'tour' back to the start point.

Part (b) About half the candidates failed to spot the 'intended' answer and spent some time trying to find a solution by trial and error.

Part (c) This process was not as widely known as the others tested on the paper and the presentation by those who did know it was also the least clearly explained.

Q10.

The use of inequalities here was where many candidates fell down. They seemed unable to apply these correctly and many wrote down equations ($=$) rather than using $<$ or $>$ signs.

Part (a)(i) was well answered, with even the weakest candidates often scoring full marks. In part (a)(ii), there were slightly fewer correct responses.

In part (b)(i), candidates who succeeded here usually went on to achieve full marks for the rest of the question, whilst those who didn't usually got no further. Most candidates immediately solved their inequality, which wasn't required by the question (although useful for later work). Part (b)(ii) was the final major hurdle. The majority of those who got to here succeeded once the correct inequalities had been identified. Almost all the candidates who got to part (b)(iii) knew how to solve the question but many unfortunately had the wrong value for x .

Q11.

This work was well known, continuing the improvement from previous series. The weakest candidates often failed to complete tours. There were very few answers presented in matrix form – a considerable proportion of these failed to provide indication of order. In part (a), a surprising number of candidates omitted to give an example, opting to give a general description or definition of a tour. Part (b)(i) was usually correct. In part (b)(ii), as with similar past questions, there was a majority view that the current figure *can* be improved rather than *might* be. Some still failed to pick up the 'tour' mark. There were many correct responses to part (c) of the question.

Q12.

Part (a) was usually answered correctly by candidates.

Part (b) was more difficult. Many candidates seemed to feel the word 'interpretation' necessitated a verbal answer and a few then completely missed the point. Use of

'between' lost the mark, as it was deemed too vague.

Part (c) did highlight the better candidates. There were many '35's for *CD* in part (c)(i) and after that there was much confusion between diagram and table. This was again not helped by generally poor presentation. Many candidates appeared to leave their question numbering for parts (c)(ii) and (c)(iii) deliberately vague.

Q13.

Overall this question discriminated well.

Part (a)(i) was almost always correct and most candidates also got part (a)(ii) correct. The most common errors in part (a)(ii) were simply to add numbers and not indicate the route clearly, or to omit the return to the start. Part (a)(iii) differentiated with a number scoring both marks and very few scoring neither.

In part (b), there was an unwillingness on the part of some candidates to explain which numbers related to which edges, and a large number of candidates produced tours without *B*. A few candidates simply found a spanning tree. Surprisingly few seemed aware of the properties of the method which were sought in part (b)(ii). A minority of those who did wrote them down even though they clearly made nonsense of the answers offered for the previous part.

Part (c) allowed successful candidates to show their understanding of travelling salesman problems. Again it was surprising what some candidates were prepared to draw and write in the light of their previous answers.

Q14.

Parts (a) and (b) were well answered, although some candidates confused 'From' and 'To' in the table. Part (c) was usually well answered; however, some candidates worked on the matrix and did not make their tour clear. Candidates are advised to list their tour. Part (d) discriminated between candidates: 66 was a common wrong answer, indicating a lack of understanding of upper bounds.

Q15.

Part (a)(i) was usually well done, although a few failed to return to *G*. However, part (a)(ii) was not done well by most, and several who might have known what to do failed to earn full marks because their method was simply not clear. Future candidates should study the mark scheme for this part to ensure they are able to do themselves justice. There were many nonsense answers to part (a)(iii), after wrong answers to previous parts, appearing to imply that a minimum can be greater than a maximum!

Parts (b)(i) and (b)(ii) were often given in reverse order, and parts (b)(iii) and (b)(iv) were poorly

answered. $\frac{n!}{2}$ was a more popular answer than the correct one.

Q16.

This question discriminated between candidates. A significant number of candidates were unable to handle a directed network and many failed to score the marks in part (a). Part (b) was beyond the majority of candidates as they merely quoted a general definition of a Hamiltonian cycle.

Part (c)(i) was well answered, although some candidates failed to return to S. Part (c)(ii) was a standard piece of bookwork but it was still poorly answered. Definitions, and understanding, of upper and lower bounds are essential. Only the most able of students was able to achieve the correct answer in part (iii).

Q17.

Part (a) of the question was poorly answered, again showing a general lack of understanding of the theory of upper and lower bounds. There were a surprising number of candidates who failed to complete the table correctly in part (b)(i). The rest of the question was well answered by the majority of candidates reflecting their ability to apply the relative algorithms.

Q18.

Candidates scored well on both parts of (a). A small number of candidates do not return to the start vertex when applying the nearest neighbour algorithm. Part (b) discriminated between the candidates with only the better scoring full marks. Centres should note that a lower bound is a minimum spanning tree + the shortest two edges from the deleted vertex. It is not allowable to use twice the shortest edge. A significant number of candidates failed to score the mark in part (c) as they wrote their answer as a strict inequality. The fundamental principles of upper and lower bounds are still not appreciated.

Q19.

Candidates scored well on both the first part and the nearest neighbour algorithm. Part (b) was reasonably well answered. It is clear that candidates can find lower bounds, however candidates must realise that the method involves adding the 2 shortest edges to a minimum spanning tree, and not twice the shortest edge. Part (c) showed a lack of understanding of lower bounds.

Q20.

Candidates scored well on both part (a)(i) and (a)(ii). Parts (b)(i) and (ii) produced a very mixed response with a significant number of candidates wanting to use a tour instead of a minimum spanning tree. Part (c) discriminated well; some candidates chose 32 as the best lower bound rather than the correct answer of 36. Also a number of candidates gave their answer as strict inequalities. The fundamental principles of upper and lower bounds are often not appreciated.

Q21.

Candidates scored well on both the first part and the nearest neighbour algorithm. Part (b) produced very few correct solutions with candidates incorrectly stating that the upper bound can be improved on. Part (d) was better answered than in previous series. It is clear that candidates can find lower bounds. Part (e) required candidates to relate the 20 to their answer in part (d). Some candidates ignored this and gave long explanations, not relevant to the question, as to why 20 was impossible. Centres should encourage candidates to consider final parts of questions as relating, in some way, to earlier work.

Q22.

This question was answered better than expected. Many candidates were able to find the upper and lower bounds in this travelling salesman problem. The difficulty came in part (c), where they showed a poor understanding of these upper and lower bounds. Many of the weaker candidates did not obtain the correct minimum spanning tree in part (b).

Q23.

Candidates performed well on the nearest neighbour algorithm, but in part (a)(ii) a common wrong answer was that this tour could be improved upon. Candidates should have realised that an upper bound is firstly a tour and secondly that it *may* be improved on.

Part (b) discriminated between the candidates, with only the best scoring full marks.

A significant number of candidates failed to score the mark in part (c) as they wrote their answer as a strict inequality. The fundamental principles of upper and lower bounds were not fully appreciated.

Q24.

All candidates were able to score on this question; however a surprising number of candidates failed to find the minimum distances between towns in part (a). Part (b) was well answered, with the majority of candidates obtaining 470. Part (c) was poorly answered with many scripts merely rewriting the order *CPTAEC* with the names of the respective towns. Candidates must be aware that the conversion of a practical problem to a classical problem involves the interpretation of a minimum Hamiltonian cycle to the original practical problem.

Q25.

Part (a) was well answered. Many candidates listed all the possible tours in part (a)(ii). There were very few correct answers to part (b), although many candidates earned one mark by realising that a factorial sign was required in the answer.

Q26.

Some candidates did not realise that the table was not symmetrical, and although they were able to obtain a tour, they did not obtain the answer of 85 in part (a). A significant number of candidates failed to complete a tour and gave the answer of *S, P, O, B, L, R*. This mistake was then repeated in part (ii).

Part (b) was the worst answered part question on the paper, despite it being a standard bookwork definition.

Candidates need to know the conditions for both upper and lower bounds.

Candidates were able to attempt part (c), but the majority of candidates obtained an answer of 107. A worrying point was that candidates seemed unconcerned that this upper bound was greater than the answers found in part (a).

Q27.

Candidates were able to find upper and lower bounds to this travelling salesperson problem. The weakness was in their understanding of upper and lower bounds. There were very few correct answers to part (a)(i).

Q28.

A significant number of candidates scored the full 10 marks in this question. However, common weaknesses were not being able to apply the nearest neighbour algorithm and simply using a tour in alphabetical order. In part (b), some candidates realised that

Manchester City had to be deleted but found a tour from Burnley back to Burnley and then added 27 to this answer. In part (c), a number of candidates wrote strict inequalities, which is incorrect as values found in part (a) and part (b) are possible upper and lower bounds.

Q29.

Although part (a) of this question was a straightforward application of Prim's algorithm, many candidates failed to score full marks. This was due to using either Kruskal's algorithm or no algorithm at all, merely showing their final spanning tree. A significant proportion of candidates who used Prim's algorithm made careless errors and gave an incorrect final tree. Part (b)(i) was a routine application of finding a lower bound, and candidates who made a mistake in the earlier part of the question could still obtain full marks in this part. Part (b)(ii) was answered poorly. The vast majority of candidates stated that the graph was not Eulerian due to the vertices having odd order, when the question required an acknowledgement that a Hamiltonian cycle was impossible.

Q30.

Although this question was an algebraic application of a travelling salesman problem, candidates were led through the question in stages. Nearly all candidates scored some marks on this question and the candidates who followed the instructions carefully scored full marks.

Q31.

Candidates used the insert as a working document and only a few chose to draw out the network in full. The question exposed some fundamental flaws in candidates' knowledge with a significant number using the nearest neighbour algorithm in part (a) and finding double the minimum connector in part (b). Part (c) tested the understanding of the 'travelling salesman' problem and proved to be a good discriminator. Candidates were still able to score the majority of marks in this part even if they made a numerical mistake in calculating the remaining spanning tree.

Q32.

Candidates had been prepared well for travelling salesman problems. Most scored well on this question but very few scored full marks. There were many vague answers and simple arithmetic errors in the responses to part (a)(ii). In part (c) many candidates included A and others found a cycle from B to B and then added the two shortest distances from B to this loop.

Q33.

Candidates could use any suitable algorithm in the first part of the question. There were a number of careless mistakes in finding the minimum spanning tree. Part (c)(i) proved to be demanding as candidates could not find the quickest route returning to town Y after going to town X . The last two parts of the question were answered well.

Q34.

Candidates could have used any suitable algorithm in the first part of the question. They should then have realised that an immediate upper bound would be twice the minimum spanning tree. The explanation as to why this may be considered to be an upper bound was poorly answered. Candidates must appreciate the significance of upper and lower

bounds in the context of travelling salesman problems. The last two parts of the question were accessible to all candidates.

Q35.

The majority of candidates were able to score a substantial number of marks on this question but full marks were rare.

Q36.

The syllabus includes shortcutting of minimum spanning trees as one method for finding lower bounds in travelling salesman problems. There are a number of different methods for finding upper and lower bounds in problems of this type and candidates are expected to be familiar with them all.

Q37.

This was the first time that a question on scheduling had been set. The question was very structured to assist candidates. Those who followed the instructions scored highly on this question. This question proved difficult for a number of candidates who were unsure about the nearest neighbour algorithm and proceeded to find four **lower** bounds by deleting one of the colours in turn.

Q39.

Part (a) was well done but, surprisingly, the majority of candidates seemed to struggle with the Chinese postperson problem in part (b).

Q41.

Practically all candidates made a reasonable effort at applying Prim's algorithm to the table, and most did so without error. Illustrating the minimum connector caused no problems but success at spotting two paths necessary to complete a round tour was mixed. Of the acceptable answers, perhaps the neatest is to open *BC* and *DE*, but very few candidates spotted that. An easy final mark was lost when candidates simply added the two extra lengths to their minimum connector's length, whereas they were actually asked for the length of the round tour.

Q42.

Again this was very well done. Candidates displayed a good working knowledge of the Chinese postperson algorithm and the nearest neighbour algorithm, and in the final part they applied their results correctly.

Q44.

Once again candidates handled the nearest neighbour algorithm and Kruskal's (or Prim's) algorithm extremely well. Even the harder part (c) was done well by a lot of candidates who recognised that, in this example, the minimum connector plus the two shortest edges from *D* did indeed fit together to form a Hamiltonian cycle.